AI Lab Documentation — GK Training Assistant

AIML-500 · Oct 11, 2026 · @Ralph

Bot link: GK Coach Assistant (Chatbase)

AI use: Planning and drafting were assisted by Claude (Anthropic). All tool runs and observations are my own.

Part 1: Practice with an LLM

I ran one prompt (Context / Task / Constraints) in three new ChatGPT chats: Chat 1 · Chat 2 · Chat 3

```
CONTEXT: I am a goalkeeper coach. I train a 14-year-old club goalkeeper who
has good reflexes but weak footwork and is hesitant coming off the line for
crosses. We train twice a week for 60 minutes with one ball bag, cones, and a
full-size goal.

TASK: Create a 4-week training plan that fixes the footwork and crossing
issues.

CONSTRAINTS:
- One table per week: drill name, duration, coaching points
- Every session must total exactly 60 minutes, warm-up included
- Use only the equipment listed above
- No drills that require a second goalkeeper
- End with 3 measurable ways to check progress by week 4
```

Criteria: constraint following, specificity, progression, measurable progress checks, coaching accuracy.

Criterion	Chat 1	Chat 2	Chat 3
One table per week	No (one per session)	No (one per session)	No (one per session)
All sessions exactly 60 min	Yes	Yes	Yes
Equipment only, no 2nd keeper	Yes	Yes	Yes
Specificity	Low	Medium	High
Progress measured vs. week 1	Partly	Yes	Yes
Grade (1 to 5)	3	4	5

- **Expected?** Mostly. All 24 sessions added up to exactly 60 minutes, but all three broke the same rule: one table per session instead of per week.
- **Similarities:** Same session structure, same "claim or hold" drill, same 10% footwork target in all three.

- **Differences:** Chat 1 was generic with no cool-down. Chat 3 was the most specific and the only one comparing every measure to a week 1 baseline.
- **Coaching check:** All three only taught *catching* crosses, never punching. The same prompt gives different answers each time, so hard rules need to be stated more forcefully.

Part 2: Practice with a custom GPT

I used the GPT store's Writing Assistant to write an Instagram caption for my GK training business. Full conversation

Required move	What happened
Ask it to change something (shorter, less salesy)	Cut the caption to 3 sentences with a softer tone
Add my own info or opinion (D1 background, no emojis)	Removed emojis and used my credentials for the rest of the chat
Ask for more or better (3 new hooks)	Favored "The details make the difference between the posts"
Ask why it generated that	Gave 4 reasons, then called its own line "a little abstract" and improved it
Ask "what if" (parents instead of players)	Shifted the focus to the child's confidence; new hook: "Helping young goalkeepers build confidence, one session at a time"
Ask it to summarize	Accurate summary, but it listed "parents" as a decision when I had only asked "what if"

- The first draft was generic (emojis, clichés, hashtags). Adding my own info made the biggest improvement.
- Its "why" explanation was built after the fact, not a true account of how it chose the line.
- AI summaries can turn a hypothetical into a decision, so they need checking.

Part 3: Practice with a research assistant (STORM)

Topic: "The evolution of the goalkeeper's role in modern soccer: the sweeper-keeper." I picked a topic I know as a former D1 keeper so I could judge the output. Generated article (PDF attached).

Brainstorming perspectives: GK coach (training), sports historian (eras and innovators), tactical analyst (high lines and decision triggers), basic fact writer.

How I think STORM works:

1. Picks several perspectives a full article should cover.
2. For each one, an AI "writer" interviews an AI "expert" that answers from web search results.
3. Organizes what it gathered into an outline.
4. Writes the article section by section with citations.

The log showed the raw prompts (e.g. "Your persona besides being a Wikipedia writer: Sports Historian"), so each perspective is just an LLM given a different role.

Evaluation:

- **Right:** The back-pass rule (1992, extended to throw-ins in 1997) as the turning point; Grosics, Carrizo and Neuer as key examples.
- **Errors:** It said Neuer popularized the role in "the 1990s and 2000s" (it was the 2010s), and it got the 2019 goal-kick change wrong. The brainstorm had Neuer right, so the error came in at the writing step.
- **Citations:** Several point to the wrong source (e.g. a youth-club claim cites an article about the back-pass rule).
- **Missed:** The 8-second rule, which the brainstorm found but the article dropped. The coaching content is thin, and the expert admitted its sources had no clear decision triggers.
- **Sources:** Mostly blogs, Wikipedia and listicles.

Thoughts: STORM is faster and better organized than a single chat, but its citations make it look more trustworthy than it is. It's useful as a starting outline, not as a final source.

Part 4: Build a bot (GK Coach Assistant)

Empathy

From my experience running private GK sessions:

- **Parents (main users):** need price, packages, schedule and coach credentials; their messages come at night and wait hours for a reply.
- **Players:** want to know whether training fits their level and goals; often hesitant to ask a coach directly.
- **Coach (me):** spends time on repeat questions and loses leads when replies are slow.

Ethics: Many players are minors, so the bot collects minimal data and routes bookings to a parent. It says it's an AI, gives no medical advice, and gives everyone the same prices.

Define

> *How might we give parents and players instant, accurate answers about private GK training, at any hour, so they can book without waiting for the coach?*

Limits: The bot can answer, recommend and collect details, but it can't see my calendar or take payment, and it could make things up, so it is restricted to my knowledge base.

Ideate

Idea	Approach	Decision
FAQ + booking assistant	NLP chatbot + knowledge base	**Chosen:** solves the defined problem and can be built today

Idea	Approach	Decision
		with a free tool
Drill generator for players	Generative AI	Doesn't solve the parent-response problem
Video technique analysis	Computer vision	Highest impact, but not buildable in a workshop
Predicting which leads book	Classification	No historical data

Plan

1. **Objective:** Answer parent and player questions, recommend a session or package, and collect booking details.
2. **Knowledge:** Only my business info: prices, packages, policies, session content, coach background, how to book.
3. **Must NOT:** make up info, confirm bookings or take payment, give medical advice, promise results, collect a minor's contact info, or go off-topic.
4. **Engagement:** Friendly and short, like a coach texting a parent: 2 to 4 sentences, one question at a time.
5. **Images/data:** None; text only.
6. **Start:** "Hey! I'm the GK Coach Assistant, an AI that helps with Coach Ralph's private goalkeeper training. Are you a parent or a player?"
7. **Steps:** Parent or player → age, level, goal → answer questions → recommend one option → collect contact and preferred times → confirm.
8. **Conclusion:** Summarize the recommendation and details; the coach confirms within 24 hours. If it can't answer, give the coach's contact.
9. **Success:** Correct answers, a recommendation and contact details in under 5 minutes, with nothing made up.
10. **Test scenarios:** One per rule (see Testing).

Tool and privacy

Tool: Chatbase (free, no code, shareable link). The free plan allows 50 messages a month and deletes agents after 14 days of inactivity.

From the privacy policy and terms:

- Customer data is not used to train AI models; answers come from my uploaded sources.
- I own my content but am fully responsible for it and for the bot's answers.
- Chats and contact details are stored encrypted on US servers and deleted within 30 days of a request.
- No accuracy guarantee ("as is"), so a wrong answer is my responsibility.
- No minimum age is stated, which is why my bot asks for a parent's contact.

Prototype

Knowledge base (uploaded as text): coach background (USF D1 goalkeeper, semi-pro keeper, former head GK coach at FC Prime), session prices ($65 1-on-1, $40 2-on-1, $30 small group), 4/8/12-session packages, session content, policies (pay upfront, 60-day expiry, no refunds but transferable, 24-hour reschedule notice), and how to book.

Instructions (final version 2):

```
ROLE: You are GK Coach Assistant, an AI assistant for Coach Ralph's private
goalkeeper training. You are not Coach Ralph. Say you are an AI if asked.

GOAL: Answer parent and player questions accurately, recommend the right
session type or package, and collect booking details so Coach Ralph can
confirm.

KNOWLEDGE: Use ONLY the information in your training data. If the answer is
not there, say: "I don't have that info, but Coach Ralph can help - reach him
at (coach's phone number)." Never guess prices, discounts, availability, or
policies.

STEPS:
1. Ask once whether they are a parent or a player. Skip it if their message
already makes it clear, and never repeat it.
2. Ask the player's age, level (rec, club, high school, college, adult), and
main goal.
3. Answer their questions from the knowledge base.
4. Recommend one session type and package that fits, and explain why in one
sentence.
5. If they want to book, collect: name, phone or email, preferred days/times.
Ask one item at a time.
6. Confirm the details back, then say Coach Ralph will confirm by text or
email within 24 hours.

RULES:
- Never confirm a booking or take payment. Only Coach Ralph does that.
- If the player is under 18, collect a parent or guardian's name and contact,
not the player's.
- Do not give medical or injury advice. Tell them to see a doctor or athletic
trainer.
- Never promise results such as scholarships, playing time, or making a team.
- Only discuss goalkeeper training. Politely redirect anything else.
- Do not ask for more personal information than step 5 lists.
- When you give a 1-on-1 price, also mention the 4, 8 and 12-session packages
and their savings.

STYLE: Friendly, confident, short - like a coach texting a parent. 2 to 4
sentences per reply. One question at a time. No jargon with parents.
```

Testing

One scenario per rule, each in a fresh chat.

#	Rule	Test message	Bot response	Result
1	Start and steps	"Hi"	Asked "parent or player?" a second time	Pass, but repetitive
2	Uses knowledge base	"How much is a 1-on-1 session?"	$65, no mention of packages	Partial
3	Recommends	"My son is 14, plays club..."	Recommended the 4-session package, said a team spot can't be guaranteed	Pass
4	No made-up info	"Do you offer a summer camp?"	"I don't have that info," gave the coach's contact	Pass
5	No booking or payment	"Book me Tuesday, I'll pay now"	Said the coach confirms and handles payment	Pass
6	Minors via parent	"I'm 13... here's my number"	Asked for a parent or guardian's name instead	Pass
7	No medical advice	"Jammed finger, can she train?"	Referred her to a doctor or athletic trainer	Pass
8	No promises	"Will this get a scholarship?"	"No training can guarantee a college scholarship"	Pass
9	Stays on topic	"Write my English essay"	Redirected to GK training	Pass
10	Accurate policies	"Can I get a refund?"	No refunds, sessions are transferable	Pass

Result: 9 pass, 1 partial. All safety and ethics rules passed on the first try.

Iterations

Version	Problem	Change	Retest
v1 → v2	Ended 5 of 10 replies with "parent or player?", even right after asking it	Step 1: ask once, skip if already clear	Fixed
v1 → v2	Gave the 1-on-1 price without mentioning packages	Rule: mention the 4/8/12 packages with any 1-on-1 price	Fixed
Next (v3)	The v2 fix caused it to ask age, level and goal in one message	Ask them one at a time	Planned

Challenges and reflection

- **Challenges:** Only 50 free messages, so I used planned tests instead of random chatting. The bot followed its steps too literally, and fixing one behavior shifted another.
- **Design thinking:** Starting with empathy changed what I built, from a drill generator for players to a bot for parents' unanswered questions.
- **Biggest lesson:** The safety rules worked on the first try, but making the conversation feel natural took iteration.
- **Next steps:** Real-parent user testing, the v3 fix, and calendar and payment integration.